Solution Requirements

Team ID: SWTID-2026-6771
Project Name: Credit card Approval Prediction

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1. Functional Requirements

S.No	Requirement Category	Requirement Description	Priority
1	User Input	Allow users to enter applicant details through a web form.	High
2	Input Validation	Validate all mandatory fields and data formats before submission.	High
3	Data Processing	Convert user input into the format required by the ML model.	High
4	Prediction	Predict whether the credit card application is Approved or Rejected using the Random Forest model.	High

5	Result Display	Display the prediction result to the user in a user-friendly format.	High
6	Model Loading	Load the trained machine learning model during application startup.	High
7	Error Handling	Display appropriate error messages for invalid inputs or system errors.	Medium
8	User Interface	Provide an easy-to-use and responsive web interface.	Medium

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2. Non-Functional Requirements

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system should generate credit card approval predictions quickly after the user submits the application.	Prediction result displayed within 2–3 seconds.
2	Scalability	The system should support multiple users accessing the application simultaneously without significant performance degradation.	Supports 100+ concurrent users with stable performance.
3	Security & Data Privacy	The system should protect applicant information and validate all user inputs to prevent unauthorized access or invalid data.	HTTPS deployment, server-side input validation, and secure storage of the ML model.
4	Reliability & Availability	The application should provide consistent predictions and remain available during normal operation.	99% uptime with consistent prediction results for identical inputs.

5	Usability & Accessibility	The application should have a simple, responsive, and easy-to-use interface for all users.	User can complete an application in under 5 minutes; compatible with desktop and mobile browsers.
6	Maintainability & Portability	The system should allow easy model updates and deployment on different platforms without major code changes.	Model can be replaced by updating the .pkl file; deployable on Render, Railway, or any Flask-supported cloud platform.